Extreme Mechanics Letters 84 (2026) 102471

Contents lists available at ScienceDirect

Extreme Mechanics Letters

journal homepage: www.elsevier.com/locate/eml

A two-layer electrostatic film actuator with high force density and integrated brake☆

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ARTICLE INFO

Keywords:
Electrostatic actuator
Soft robotics

ABSTRACT

Robotic systems driven by conventional motors often face challenges such as large mass, complex control algorithms, and the need for additional braking mechanisms, which limit their applications in lightweight and compact robotic platforms. Electrostatic film actuators offer several advantages, including thinness, flexibility, lightweight construction, and high open-loop positioning accuracy. However, the force density of conventional actuators in air still needs improvement, particularly for the widely used three-phase electrode design. To enhance the output performance of actuators, this paper presents a two-layer electrostatic film actuator with an integrated brake. By alternately distributing electrodes on both the top and bottom layers, a smaller effective electrode pitch is achieved under the same fabrication constraints, resulting in a force density of approximately 241 N/m^2, representing a 90.5% improvement over previous three-phase actuators operating in air. Furthermore, its integrated electrostatic adhesion mechanism enables load retention in braking mode. Several demonstrations, including a tug-of-war between a conventional single-layer actuator and the proposed two-layer actuator, a carried-mass transport demonstration, a one-degree-of-freedom robotic arm, and a dual-mode gripper, were conducted to validate the actuator's advantageous capabilities in both actuation and braking modes.

1. Introduction

With the development of industrial automation and intelligent manufacturing technologies, the demand for robotic systems in human society has been steadily increasing. However, most robotic systems are based on magnetic motors combined with rigid transmission mechanisms, resulting in substantial mass [1–3] and limiting their applications in human–machine interaction and miniature robotics. Furthermore, to withstand static loads while maintaining the desired position, magnetic motors typically require external braking components [4–6] or closed-loop control systems [7–9], increasing the complexity of the system. Soft actuators, such as dielectric elastomers [10,11], shape memory alloys [12,13], and pneumatic actuators [14–16], have been widely applied in grippers, wearable devices, and miniature robots [17,18] in recent years. However, their viscoelastic materials often lead to limited accuracy and long response times [19,20].

Electrostatic film actuators (EFAs), driven by electrostatic forces generated between microscale comb electrodes embedded within the polymer insulation film [21,22], offer advantages including thinness, flexibility, lightweight construction, and high open-loop positioning accuracy. In addition, they can provide sufficient actuation velocity and force [23], making them suitable for applications in human–machine interaction [24,25] and bio-inspired robotics [26–28]. Due to their operation under high driving voltage amplitude but relatively low current, EFAs also exhibit resistance to strong magnetic interference [29,30].

However, in conventional flexible printed circuit manufacturing processes, both the electrode width and spacing are constrained by the same minimum feature size. Conventional EFAs arrange all electrodes within a single electrode layer [31–33], limiting further reduction of the effective electrode pitch (the sum of width and spacing) and thus restricting the achievable actuation force [34]. Moreover, the asymmetry in the distance between the electrode layer and the surfaces results in unbalanced actuation forces on both sides of the film, which degrades performance when multiple films are stacked [22,35]. A four-phase two-layer EFA design has previously been reported to achieve higher force density and has been applied to mesoscale artificial muscles [34]. However, it requires more precise fabrication techniques and more

☆ This work was supported in part by the Natural Science Foundation of China under Grant 52275021 and 52450323, in part by the Natural Science Foundation of Guangdong Province of China under Grant 2024A1515010183, in part by 2025 Shenzhen–Hong Kong–Macau Science and Technology Program (Category C) (SGCX20250526153900001), and in part by Guangdong Provincial Research Support Program under Grant 2019QN01Z733.

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https://doi.org/10.1016/j.eml.2026.102471
Received 4 January 2026; Received in revised form 8 February 2026; Accepted 9 March 2026
Available online 10 March 2026
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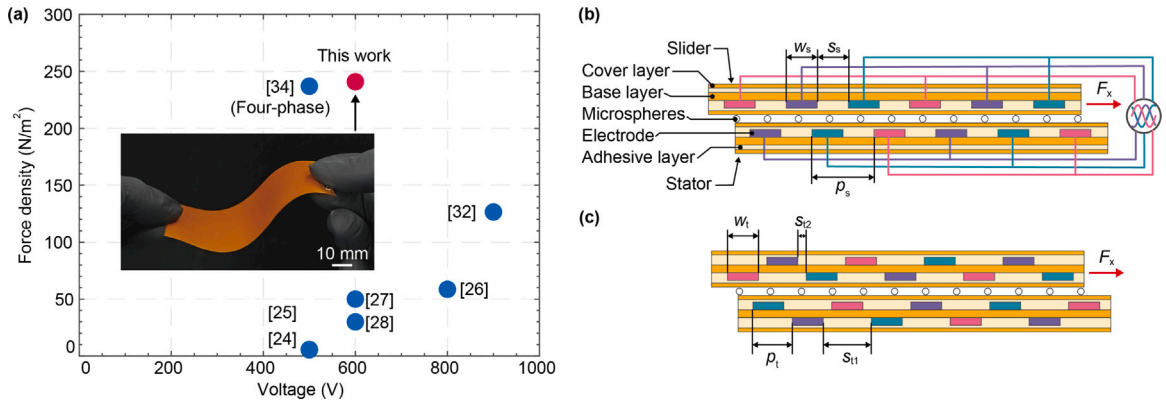


Fig. 1. The two-layer EFA. (a) Demonstration of the flexible and thin structure of the two-layer EFA, and comparison of the force density of the proposed two-layer three-phase actuator with previously reported EFAs operating in air. Principle and design of (b) conventional single-layer EFA and (c) the two-layer EFA.

driving equipment. Accordingly, the development of electrode designs capable of enhancing the actuation force density within the constraints of existing fabrication processes remains a challenge in advancing EFA performance.

The contributions of this work are as follows. In this work, we propose a two-layer EFA (Fig. 1a), where the electrodes are alternately arranged more densely across the upper and lower layers. This design enables a smaller effective electrode pitch under the same fabrication conditions, resulting in a higher force density (the output force normalized by the effective actuation area) than conventional three-phase EFAs at the same driving voltage amplitude, as shown in Fig. 1a. In addition, the larger spacing between electrodes in the same layer improves the breakdown strength compared to the four-phase EFA, achieving force density up to approximately 241 N/m^2 and allowing the actuator to operate with a carried mass of approximately 68.2 times its own weight. Benefiting from the symmetric two-layer electrode design, the actuation film can generate equal output forces on both sides, facilitating force enhancement by stacking multiple films. Furthermore, the electrode coverage ratio reaches 83.3% (with half of the electrodes located in the other layer), enabling the actuator to generate a large electrostatic adhesion force for braking functionality. Based on this two-layer EFA with integrated brake, we demonstrate its application in driving a one-degree-of-freedom (DOF) robotic arm capable of handling heavier loads during active braking. Moreover, a dual-mode gripper is developed, utilizing the EFA's actuation mode for object grasping and its braking mode for object holding.

The structure of this paper is organized as follows. Section 2 presents the working principle and design of the two-layer EFA. Section 3 describes the fabrication process and the microstructure of the actuator electrode. Section 4 details the experimental setups and evaluates the actuator's performance. Section 5 demonstrates the applications of a robotic arm and a gripper. Section 6 concludes the paper and discusses future work.

2. Actuator design and analyses

2.1. Principle and design of the two-layer EFA

The design of the slider and stator in conventional single-layer EFAs is shown in Fig. 1b. Three-phase electrodes are arranged in parallel on the base layer and electrically insulated by the cover layers through adhesive layers. In terms of electrode parameters, w_s denotes the electrode width, s_s represents the electrode spacing, and p_s, defined as the sum of the width and spacing ($p_s = w_s + s_s$), is referred to as the electrode pitch. Due to the identical minimum size limitation of the fabrication process for both width and spacing, w_s is typically designed to be equal to s_s. As a normal adhesion force is generated between the slider and the stator during operation, glass beads (with a diameter of 20 to 30 μm) are uniformly applied to the contact surfaces of the films to reduce friction. When three-phase sinusoidal voltages with initial phase differences of 120° are applied to the electrodes, a horizontal driving force F_x is generated on the slider, which can be expressed as [23]:

$$F_x = K_a v_a^2 \sin(\theta_e - 2\phi), \tag{1}$$

where $K_a = k_a A_e$ is the actuation force constant with units of N V^{-2} of the actuator. k_a is a material-dependent force coefficient with units of N m^{-2} V^{-2}. A_e is the effective actuation area of the actuator. v_a is the amplitude of the driving voltages. $\theta_e = 2\pi x_{sl}/3p$ represents the electric angle of the slider position x_{sl} along the X direction, with p denoting the effective pitch. The parameter ϕ is the excitation phase of the applied voltage.

Since EFAs operate in synchronous motion, their actuation velocity can be expressed as [36]:

$$v = 6pf, \tag{2}$$

where f is the driving frequency of the applied voltage.

The design of the proposed two-layer EFA is illustrated in Fig. 1c. The electrodes are alternately arranged on both the top and bottom layers. In this configuration, s_{t1} denotes the electrode spacing on the same layer, s_{t2} represents the adjacent electrode spacing on different layers, and the effective electrode pitch $p_t = w_t + s_{t2}$. When utilizing an identical fabrication process for both single-layer and two-layer designs, the electrode widths remain the same ($w_t = w_s$). Since s_{t2} can be adjusted without requiring higher fabrication precision, a smaller effective pitch can be achieved when s_{t2} is between 0 and s_s, resulting in p_t smaller than p_s. Meanwhile, the same-layer electrode spacing satisfies $s_{t1} = w_t + 2s_{t2}$, which is larger than s_s ($s_s = w_s = w_t$), resulting in a higher breakdown strength and thus enabling a higher maximum actuation force of the EFA.

In addition to actuation, the electrostatic adhesion capability of the two-layer EFA can be utilized for braking. During adhesion operation, all slider electrodes are connected to a square-wave AC voltage, while all stator electrodes are grounded. Under this configuration, the system can be approximated by a parallel-plate capacitor model, and the normal adhesion force can be calculated as [37]:

$$F_n = K_b v_b^2, \tag{3}$$

where K_b is the adhesion force constant with units of N V^{-2} based on the materials and dimensions of the electrostatic adhesive pad, v_b is the amplitude of the driving voltage for braking. Since the actuation force operates tangentially, the corresponding tangential adhesion force used for braking can be expressed as:

$$F_t = \mu_b F_n = \mu_b K_b v_b^2, \tag{4}$$

where μ_b is the friction coefficient between the slider and the stator.

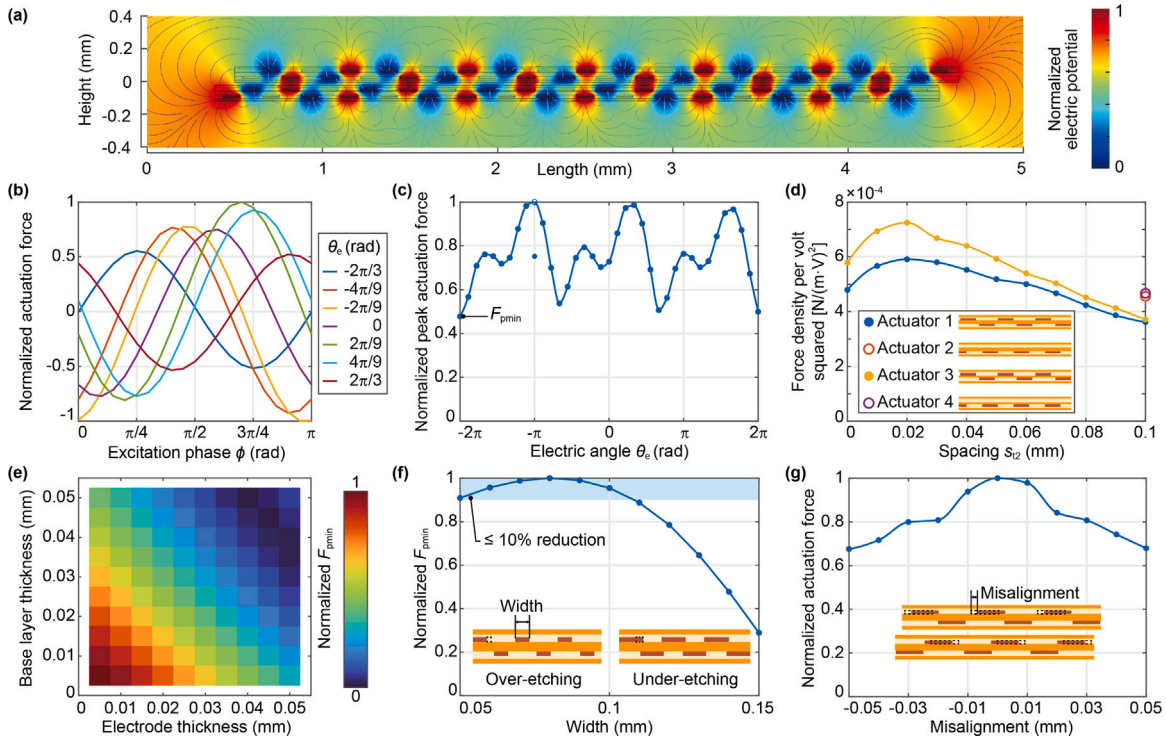


Fig. 2. Analysis of the two-layer EFA and the effects of design and fabrication parameters on the actuation force. (a) Normalized electric potential distribution of Actuator 1. (b) Normalized actuation force as a function of excitation phase under different electric angles. (c) Normalized peak actuation force versus electric angle. (d) Force density per volt squared as a function of electrode spacing. (e) A contour map of the normalized actuation force under different electrode and base layer thicknesses. (f) Effect of electrode width on the actuation force. (g) Normalized actuation force as a function of electrode misalignment.

Table 1
The parameters of four types of actuators in simulations.

Actuator	Single/two-layer	Thickness (μm)	
		Electrode layer	Base layer
Actuator 1	Two	12	25
Actuator 2	Single	12	25
Actuator 3	Two	18	12.5
Actuator 4	Single	18	12.5

2.2. Finite element analysis

Finite element analysis was employed to investigate the effects of electrode configurations and fabrication parameters on the actuation forces of EFAs. In the simulation, the slider and stator were positioned inside an air-filled enclosure, with an air gap of 20 μm and a relative permittivity of 1. The base layer and cover layer materials were set as polyimide with a relative permittivity of 3.4, while the adhesive layer was set as acrylic adhesive with a relative permittivity of 3.5. The electrode material was set as copper. The thicknesses of the cover layer and adhesive layer were set to 13 μm and 2 μm, respectively. As summarized in Table 1, simulations were carried out for four types of actuators to compare their actuation forces under different design parameters. Specifically, combinations of electrode layer thickness and base layer thickness were selected based on two kinds of commercial double-sided copper-clad laminates.

By applying driving voltages to the electrodes of each phase in the simulation, as shown in Fig. 2a, the electric potential distribution of the two-layer EFA (corresponding to Actuator 1 in Table 1, with w_t = 100 μm and s_{t2} = 20 μm) was obtained. This further yielded the relationship between the normalized actuation force and the excitation phase at different electric angles within two effective pitches (the period of the two-layer actuator spans six effective pitches), as shown in Fig. 2b. The relationship between the peak actuation force and electric angle is presented in Fig. 2c, where F_{pmin} denotes the maximum achievable normalized actuation force during continuous motion.

To identify optimal design parameters for the two-layer EFA, the influence of the spacing s_{t2} on actuation force was further investigated. In this comparison, all electrode widths were fixed at 100 μm. For the single-layer EFAs (Actuators 2 and 4), the electrode spacing s_s was set to 100 μm. The relationship between force density per volt squared and spacing s_{t2} is shown in Fig. 2d. The two-layer EFAs exhibited maximum actuation forces when s_{t2} was set to 20 μm, and were much higher than those of the single-layer counterparts (the force density per volt squared of Actuator 1 is 29.7% larger than that of Actuator 2). The analysis results for the normalized F_{pmin} as a function of the electrode and base layer thicknesses are shown in Fig. 2e, indicating that increasing either layer's thickness reduces the actuation force. Based on the materials currently available to us, we adopted Actuator 1 instead of the higher performing Actuator 3 in this work.

During the fabrication of the two-layer EFA, electrode over-etching, under-etching, or misalignment may occur due to a processing error. The effect of electrode width errors is shown in Fig. 2f. Over-etching has a limited influence, as a 50 μm reduction (corresponding to Actuator 1 with w_t = 100 μm and s_{t2} = 20 μm) in width decreases F_{pmin} by only ~10%. In contrast, under-etching is more detrimental. A 0.01 mm (10%) increase in width reduces F_{pmin} by > 10% and should be avoided. The influence of the misalignment on the actuation force was analyzed through simulation. The relationship between normalized actuation force and electrode misalignment is shown in Fig. 2g. As the misalignment value increases, the actuation force decreases significantly. Therefore, the misalignment should be controlled within 10 μm during fabrication to ensure optimal actuator performance.

3. Fabrication

The fabrication procedure of the two-layer EFA film is shown in Fig. 3a. Firstly, a double-sided copper-clad laminate (comprising a 25 μm

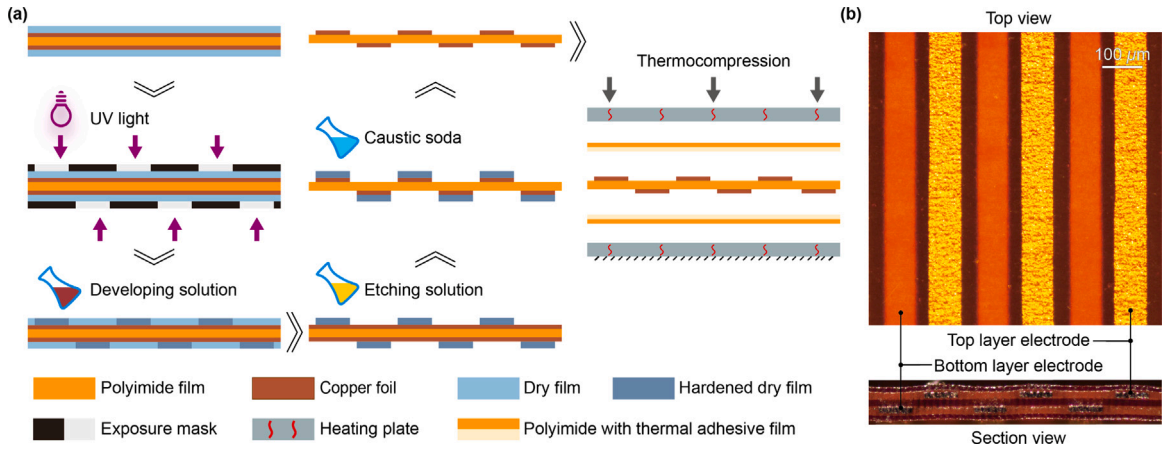


Fig. 3. Fabrication of the two-layer actuator. (a) The illustration of the fabrication process. (b) The top and sectional views of the electrodes.

Table 2
Measured fabrication parameters of the actuation films.

	w_t (μm)	s_{t2} (μm)	Misalignment (μm)
Design parameter	100	20	0
Sample 1	89.3	33.2	0.41
Sample 2	89.7	27.5	4.94
Sample 3	87.6	31.1	5.64
Sample 4	89.0	33.2	0.71
Sample 5	89.0	31.1	2.82
Mean	88.9	31.2	2.90

polyimide base layer and two 12 μm copper layers) served as the substrate for the actuator. Then, negative-type dry films were laminated to both sides of the substrate. Ultraviolet (UV) light was subsequently projected through exposure masks containing the designed electrode layouts. The exposed portions of the photosensitive dry films were hardened, forming insoluble resist layers, while the unexposed regions were removed by a developing solution (1% Na_2CO_3). The uncovered copper was then etched away using an etching solution ($CuCl_2$ and HCl), leaving the designed electrode patterns. After etching, the residual hardened dry films were stripped with a caustic soda solution (3% NaOH), exposing the patterned flexible circuits. For insulation, two coverlays (consisting of a 13 μm polyimide layer and a 13 μm thermal adhesive layer) were laminated onto both sides of the etched substrate using a thermocompression at 185 °C and 2.3 MPa for 3 h.

The final fabricated two-layer EFA film measured 100 mm×50 mm in length and width, with a total thickness of 90 μm and a mass of 0.88 g. The effective electrode area was 92.8 mm × 45.6 mm. The microscopic top view and the section view of the fabricated electrodes are shown in Fig. 3b. Inside the actuation film, the electrodes were alternately distributed on the top and bottom layers. The fabricated copper electrodes had an average width of approximately 88.9 μm, a spacing of approximately 31.2 μm, and a misalignment of approximately 2.90 μm, as shown in Table 2. These dimensions slightly deviate from the design values, which may be attributed to over-etching during fabrication. However, based on Figs. 2f and 2g, the resulting deviations have a negligible impact on actuator performance.

4. Experiment results

We experimentally characterized the velocity, lifetime, positioning accuracy, actuation force, braking force, and power consumption of the two-layer EFA. In these experiments, control signals were generated by a controller board (SCB-68 A, National Instruments) and amplified 1000 times using three high-voltage amplifiers (Model 615-3, Trek) to drive the actuator. Sensor data were collected using analog signal acquisition boards (TB-4309 and TB-4330, National Instruments).

4.1. Results of velocity, lifetime, and positioning accuracy

In the displacement measurement setup shown in Fig. 4a, the stator was fixed on an acrylic plate, while a reflector was mounted on the slider for displacement sensing via a laser displacement sensor (HG-C1200, Panasonic). In displacement tests, the voltage amplitude was set at 500 $V_{0\text{-}p}$ (zero-to-peak voltage). The displacements of the actuator under different driving frequencies are presented in Fig. 4a. As shown in Fig. 4b, the actuator velocity exhibited a linear relationship with the driving frequency, which is consistent with the theoretical model. The maximum velocity reached approximately 143 mm/s at 200 Hz. As shown in Fig. 4c, the actuator can operate for more than 10,000 cycles at a driving frequency of 25 Hz, a voltage amplitude of 500 $V_{0\text{-}p}$, and a stroke of 9 mm while maintaining stable reciprocating motion without observable degradation in displacement amplitude. To evaluate linear positioning accuracy, the slider was driven to reciprocate over a 14.4 mm stroke at a driving frequency of 20 Hz for ten cycles, as shown in Fig. 4d. According to the results and based on the accuracy evaluation method of the Physik Instrumente (PI) [38], the mean bidirectional positioning accuracy of the actuator was 0.048 mm, with a bidirectional repeatability accuracy of 0.035 mm as shown in Fig. 4e.

4.2. Results of the actuation force

In the actuation force measurement setup shown in Fig. 4f, the stator was fixed on a 3D-printed sliding rail through screws. The slider was connected to a force sensor (LSB 201, FUTEK) via a hook and a spring. During the test, the actuator was applied with sinusoidal voltages with a 10 Hz driving frequency. As the slider stretched the spring leftward, the force sensor recorded the increasing force in real-time. The maximum force value was recorded as the actuation force since the slider could no longer extend the spring at a given voltage amplitude. The relationship between actuation force and voltage amplitude is shown in Fig. 4f, which is in agreement with the finite element model. A maximum actuation force of 1.02 N (corresponding to a force density of approximately 241 N/m^2) was achieved at 600 $V_{0\text{-}p}$ voltage amplitude.

4.3. Results of braking force

The experimental setup for evaluating the electrostatic adhesion braking force is shown in Fig. 4f. The slider was connected to a force

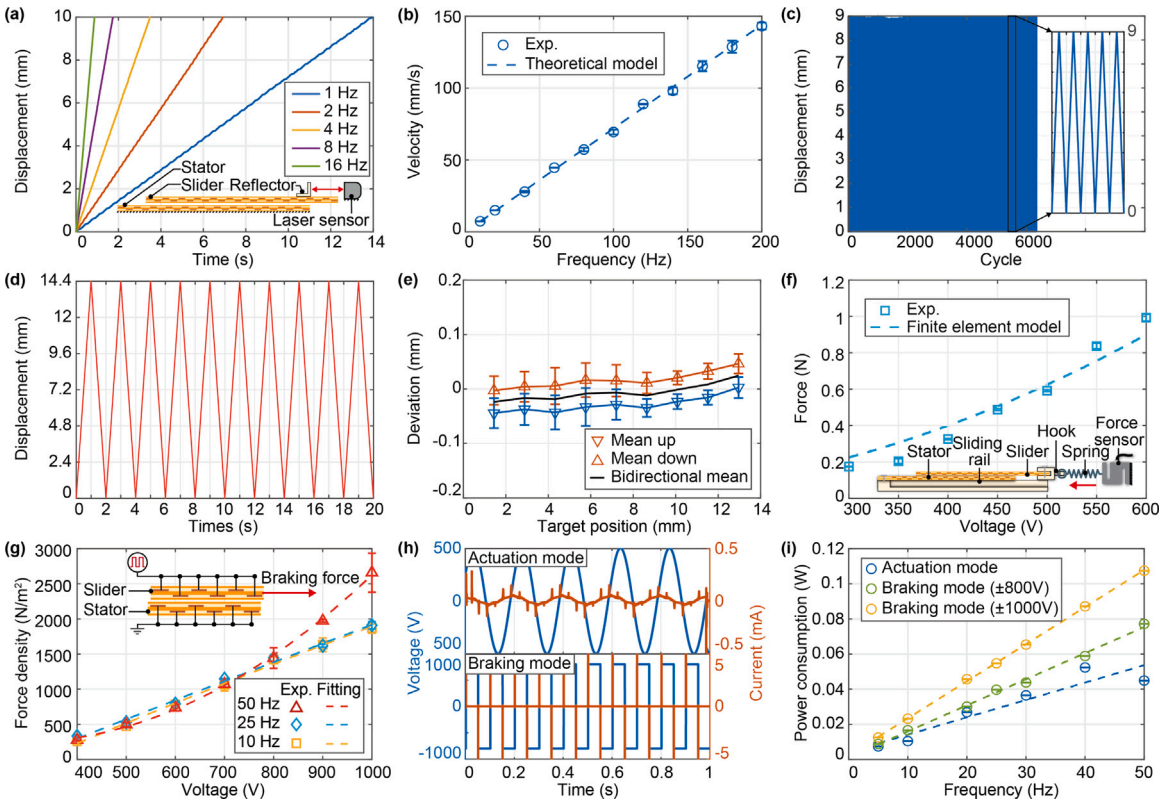


Fig. 4. Performance characterization of the actuator in actuation and braking modes. (a) Schematic of the experimental setup and displacement responses at different driving frequencies. (b) Velocity as a function of driving frequency. (c) Cycling test under a 9 mm stroke at a 25 Hz driving frequency. (d) Displacement data for 10 cycles of reciprocal motion. (e) Positioning error. (f) Schematic of the actuation force measurement setup and actuation force under different voltage amplitudes at a 10 Hz driving frequency. (g) Braking force density under different voltage amplitudes and driving frequencies. (h) Output voltage and current signals of the high-voltage amplifiers. (i) Power consumption of the actuator in actuation mode and braking mode. Error bars represent the mean ± 1 SD ($n = 3$).

sensor (LSB203, MTS) and moved at a constant velocity of 50 mm/min by a tensile test platform (Model 42, MTS). During testing, all slider electrodes were driven by a bipolar square-wave voltage while the stator electrodes were grounded, with a contact area of 90×50 mm^2. The relationships between braking force density and voltage amplitude under different driving frequencies are shown in Fig. 4g. The results indicate that the braking force densities at 10 Hz and 25 Hz were similar. However, at 50 Hz, the force density increased significantly at high voltage amplitude and reached a maximum of approximately 2667 N/m^2 at ±1000 V. It is noteworthy that repeated vertical tensile tests may cause glass beads to detach from between the two films, increasing the friction coefficient. Therefore, the braking force measured in the tensile tests may be higher than that achievable under some operating conditions.

4.4. Results of power consumption

We measured the output voltage and current of the high-voltage amplifiers when driving the actuator in both actuation and braking modes. Fig. 4h shows representative waveforms for one phase in actuation mode (500 $V_{0\text{-}p}$, 5 Hz) and in braking mode (± 800 V, 10 Hz). The resulting average power is 7.92 mW in actuation mode and 23.3 mW in braking mode under this condition. We further quantified the power consumption over different driving frequencies (both modes) and braking-voltage amplitudes (braking mode). As shown in Fig. 4i, the power consumption increases approximately linearly with driving frequency in both modes. Since braking mode exhibits higher power, the power consumption can be reduced by lowering the braking voltage amplitude and/or the driving frequency in scenarios requiring a smaller braking force.

5. Demonstrations

5.1. Tug-of-war

To visually compare the actuation forces between conventional single-layer EFA and the two-layer EFA, a tug-of-war platform was designed as shown in Fig. 5a and Movie. On the left side, a single-layer actuator with a contacting area of 70×50 mm^2 was placed, while a two-layer actuator with a contacting area of 60×50 mm^2 was positioned on the right side. The sliders of both actuators were connected via a polyimide tape. Both actuators were simultaneously driven by sinusoidal voltages at 500 $V_{0\text{-}p}$ amplitude and 5 Hz driving frequency.

At the beginning of the competition, the tape remained slack. After approximately 2 s, the tape was pulled taut, and both actuators began exerting opposing forces. Then, the single-layer actuator was pulled off-axis and lost the competition. By 9 s, the two-layer actuator had dragged the single-layer actuator out.

5.2. Carried mass operation

To evaluate the carried mass capability of the two-layer EFA, a testing platform was constructed as shown in Fig. 5b. The stator was mounted on a sliding rail, while the slider was placed on top of the stator. When driven by sinusoidal voltages at 500 $V_{0\text{-}p}$ amplitude and 30 Hz driving frequency, the slider performed a reciprocating motion, as shown in Fig. 5b and Movie for the unloaded case.

Subsequently, a 3D-printed plate (20 g) and an additional mass (100 g) were placed atop the slider, resulting in a total carried mass equivalent to approximately 68.2 times the actuator's own weight. As shown in Fig. 5b, the actuator maintained its full stroke with-

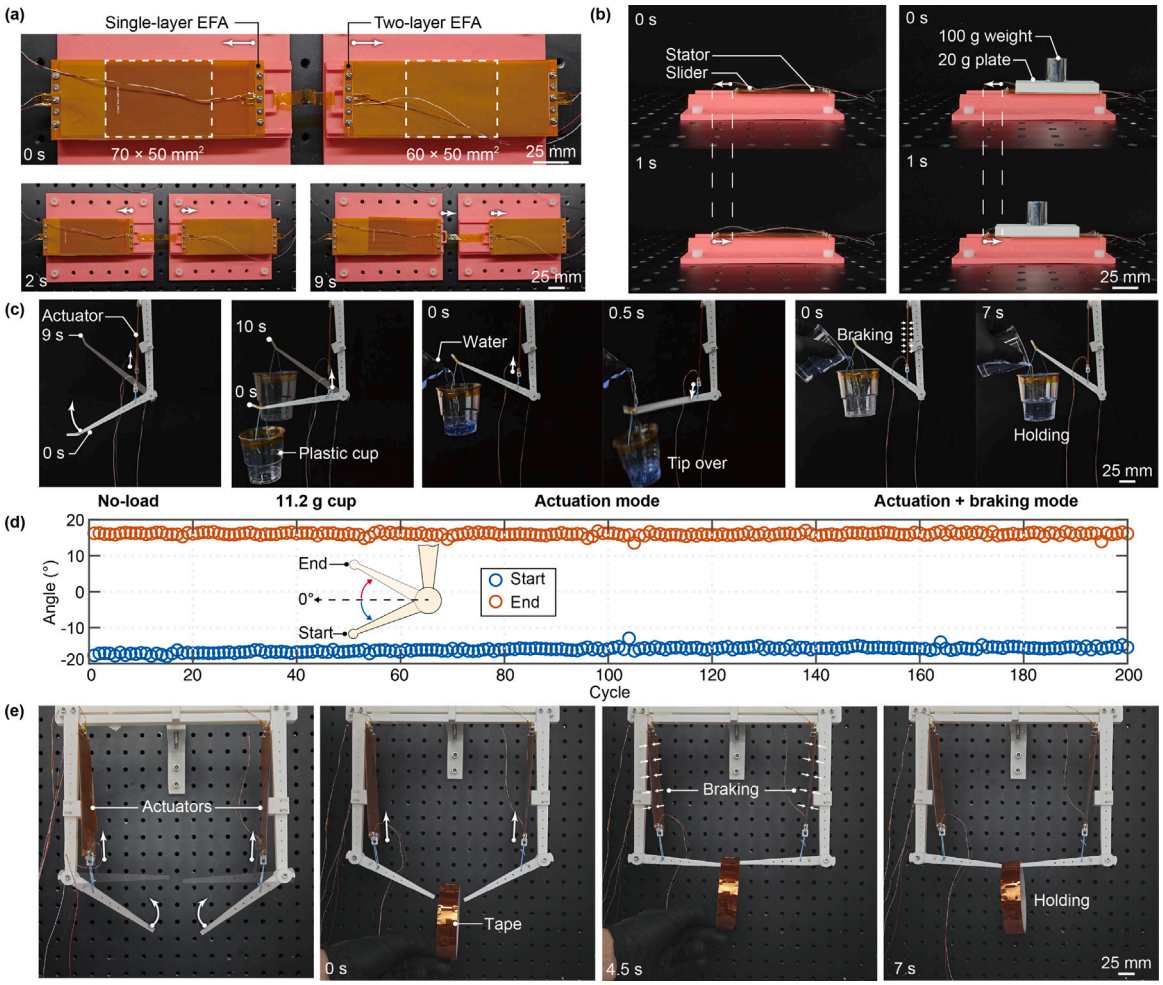


Fig. 5. The demonstrations of the actuator. (a) The demonstrations of a tug-of-war between a single-layer EFA and the proposed two-layer EFA. (b) The demonstrations of the carried mass capacity of the actuator. (c) The demonstrations of the actuator for a one-DOF robotic arm. (d) Angle of the start and end points of the arm over repeated cycles. (e) A dual-mode gripper performs object grasping through the actuation mode of the actuator, and holds the object after closure using the braking mode.

out observable degradation in motion performance under this loading condition.

5.3. One-DOF robotic arm

To demonstrate the integrated braking capability of the two-layer EFA, the actuator was implemented as both the actuation and braking mechanism for a one-DOF robotic arm. Benefiting from the force superposition capability of the two-layer design, the actuator was constructed using three stator films and two slider films. The stator was mounted on the upper arm, while the slider was connected to the forearm via a thin cable. The actuator was driven by voltages at 500 $V_{0\text{-}p}$ voltage amplitude and 5 Hz driving frequency.

As shown in Fig. 5c and Movie, the actuator successfully drove the robot arm in elbow-like rotational motion under unloaded conditions. The actuator could drive the arm for approximately 200 reciprocating cycles without obvious performance degradation (Fig. 5d). The arm was also able to lift a plastic cup with a mass of 11.2 g from a lower initial angle, as illustrated in Fig. 5c. To further evaluate the carried mass limit during dynamic motion, water was incrementally added into the cup during operation. As shown in Fig. 5c, when the added water mass exceeded approximately 20 g at 1.5 s, the actuation force was no longer sufficient to maintain motion, resulting in slipping backward and ultimately causing the cup to tip over at 0.5 s.

After reaching the target position by actuation mode, the actuator was switched to braking mode by applying a ±1000 V square-wave voltage at 50 Hz to the slider electrodes while grounding the stator electrodes. Under braking mode, as shown in Fig. 5c, the actuator was able to hold the arm position even when the carried mass exceeded 60 g at 7 s, preventing tipping and maintaining stability.

5.4. Dual-mode gripper

Based on two one-DOF robot arms, a dual-mode gripper was constructed by modifying the end-effector structures and assembling them into a gripping mechanism, as illustrated in Fig. 5e and Movie. The gripper successfully grasped a polyimide tape weighing approximately 20 g. During the grasping process, both actuators operated in actuation mode (500 $V_{0\text{-}p}$ voltage amplitude with 5 Hz driving frequency) to drive the gripper fingers inward, allowing insertion into the hollow center of the tape. After approximately 4.5 s, once the gripper fully closed, the actuators were switched to braking mode (applying ±1000 V square-wave voltage at 50 Hz to the slider electrodes while grounding the stator electrodes), thereby holding the grasped object.

6. Conclusions

In summary, a two-layer EFA with integrated braking capability was proposed and experimentally demonstrated to address the force limitations of conventional EFAs operating in air. The two-layer electrode design enables a smaller effective electrode pitch without requiring

higher fabrication precision, thereby significantly enhancing the force density. Finite element simulations were conducted to analyze the effects of design and fabrication parameters on actuator performance. Experimental results confirmed that the proposed actuator achieved high force density, precise positioning, and substantial carried mass capacity. A series of demonstrations, including tug-of-war, carried mass operation, robotic arm manipulation, and dual-mode gripping, further validated its applicability in both actuation and braking modes. Currently, the integrated braking mechanism is affected by the glass beads during operation, which may significantly reduce the braking force. Future improvements may involve encapsulation of the actuator in a dielectric liquid to ensure stable braking performance. Additionally, the present fabrication process employs a relatively thick base layer, which could be further optimized to enable the two-layer design to achieve even higher force density. This two-layer EFA provides a promising solution for lightweight robotic systems that require compact, precise actuation combined with integrated braking functionality.

CRediT authorship contribution statement

Huacen Wang: Writing – original draft, Data curation, Conceptualization. **Ting Wang:** Investigation, Data curation. **Hongqiang Wang:** Writing – original draft, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eml.2026.102471.

Data availability

Data will be made available on request.

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